

SZL Holdings — Business Proposal (v2)

Governed Operational Intelligence for Federal and State Agencies

Prepared for Empire APEX Accelerator

Refresh: May 5, 2026 — supersedes v1 (May 4, 2026)

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1. Company Overview

SZL Holdings builds governed operational intelligence platforms for regulated enterprises and government agencies. The platform bridges data ingestion, risk analysis, AI-supported decision-making, execution, and auditability into a single continuous workflow.

Every AI-assisted decision the platform makes carries a cryptographic receipt back to the inputs that produced it, the policy that authorized it, and the human who approved it where required. This is not a feature — it is a property of the runtime.

Field	Value
Legal entity	SZL Holdings, single-member, New York State
Founder	Stephen Lutar
Business size	Small business, single-founder
Primary NAICS	541512 — Computer Systems Design Services
Secondary NAICS	541511, 541519, 541611, 541690
ORCID	0009-0001-0110-4173

2. The Problem (unchanged)

Federal and state agencies are adopting AI systems at accelerating pace. Current procurement frameworks — NIST AI RMF, DoD Responsible AI Strategy, OMB M-24-10, New York S.B. 7599 — all require something most AI vendors cannot deliver: **runtime auditability at the point of decision**.

Existing vendors provide AI capabilities (inference, summarization, classification) but ship no mechanism to prove the AI was trustworthy at the moment it made a decision. The audit trail, when it exists, is reconstructed after the fact. The human oversight, when it is claimed, is a checkbox, not a runtime gate.

3. The Solution: A Governed Command Layer

Five continuous stages — Ingest, Score, Decide, Act, Verify — formalized as a bounded loop with a published convergence theorem (Section 6) and shipped as an open-source runtime.

The convergence properties of this loop are formalized in three peer-style papers published on Zenodo:

Paper	Title	DOI / Release
v3	The Loop Is the Product: Measuring Bounded Recursion as a System Primitive for Auditable AI	DOI 10.5281/zenodo.19944926
v9 (NEW)	The Lutar Invariant Family $v1 \rightarrow v7 \rightarrow \Omega$: From Three-Term Foundation to Bianchi-Closed Fiber Bundle	GitHub Release paper-v9-1.0.0
v10 (NEW)	The Audit Closure Operator Λ_{10} : Formalising the Implementation Contract of the Lutar Family	GitHub Release paper-v10-1.0.0

4. Platform Primitives (unchanged)

Six primitives, identical across every vertical:

- Outcome Graph · Proof Chain · Covenant Policy · Decision Simulation · Workflow Engine · Event Fabric (PRISM Bus)

5. Live Product Surfaces — refreshed

The platform now runs as **eight customer-facing product surfaces** orchestrated by A11oy.

Product	Domain	Government Relevance
A1loy	Orchestration + Decision Intelligence + Trust Plane	AI governance, audit infrastructure, proof distribution
Sentra	Cyber Resilience Command + governed adversary loop	SOC ops, CISA / Zero Trust alignment
Terra	Real Estate Intelligence	State property portfolio management
Vessels	Maritime Intelligence (live AIS feed)	DHS / Coast Guard MDA
Counsel	Legal Matter Command	Agency legal ops, traceable matter management
Amaru	Convergent Reverse-ETL + 10 original innovations	Cross-system data lineage
Carlota Jo	Concierge Advisory	Premium service brand
ROSIE (NEW)	Unified Decision Fabric	Operator surface for CPS payloads

6. Public Proof Artifacts — refreshed

Artifact	Status
Ouroboros Runtime	github.com/szl-holdings/ouroboros v6.2.0, full test suite, dependabot, CodeQL, SECURITY.md
Ouroboros Thesis v3	paper-v3-2.0.0 , DOI 10.5281/zenodo.19944926 , CC-BY-4.0
Ouroboros Thesis v9 (NEW)	paper-v9-1.0.0 , 17 pp, May 5 2026 — Zenodo deposit re-fired
Ouroboros Thesis v10 (NEW)	paper-v10-1.0.0 , 11 pp + Appendix A essay + Lutar-family one-pager, May 5 2026
Live LaaS API	<code>POST /api/ouroboros/lutar/v10</code> — runs the v10 audit closure operator against the live shipping repo on every test run

The v9 → v10 sequence is a meta-invariant on the implementation chain: v10 introduces no new physical L-term. Its only purpose is to certify, layer by layer, that every formula in v9 executes against the live repo. The platform audits its own thesis.

7. Covenant Proof Standard (CPS) — new in v2

CPS is a payload-and-receipt protocol for governed cross-vertical workflows, exposed as a first-class API. Three flagship payloads ship with the standard; per-lane payloads are rolling out across Vessels, Terra, Counsel, and Carlota Jo.

Endpoint	Behavior
GET /api/cps/payloads	List registered payloads
POST /api/cps/runs	Execute a payload run against a tenant
POST /api/cps/runs/:id/approve	Tier-checked human approval gate
POST /api/cps/runs/:id/rollback	Verified rollback to prior state
POST /api/cps/payloads/:id/maturity	Promote/demote payload maturity mode

A live agent gateway (artifacts/api-server: agent-gateway) sits in front of every agent action and enforces OPA bundle policy at the runtime boundary. Agent Zero Trust (/agent-zero-trust) is the operator-facing surface for that gate.

8. Government Standards Alignment

NIST AI RMF

Function	SZL Coverage
GOVERN	Validator registry, loop policy, operator modes, CPS approval tiers
MAP	Domain pack router, task-type routing, payload registry
MEASURE	Delta and consistency scores per step, uncertainty quantification, mirror-eval, reward-hacking guardrails (Argo)
MANAGE	Human approval gate, halt conditions, replay verification, CPS rollback

DoD Responsible AI Tenets

Tenet	Coverage	Status
Responsible	Human approval at risk tier 3/4, CPS tier check, validator hard stops	Covered
Equitable	Bias-testing plan in development	In Progress
Traceable	Full trace runtime, append-only logs, CPS receipts	Covered
Reliable	Golden runs, replay verification, Lutar v10 audit closure	Covered
Governable	Approval gate, halt conditions, agent-gateway OPA enforcement	Covered

NY State AI Procurement Climate

- NY DIGIT vendor standards for AI adoption across 50+ state agencies
- S.B. 7599 AI Transparency Law — public disclosure of AI tools and bias mitigation
- OGS centralized contract vehicles for AI governance

9. Target Sectors

Sector	Use Case	Vehicle
DoD Responsible AI	Runtime auditability for AI-in-the-loop decisions	SBIR Phase I, subcontract
DHS / CISA	AI governance under M-24-10 inventory requirements	Sources-sought response
NIST	Reference implementation for runtime trust scoring	Direct engagement
GSA Schedule 70 / MAS IT	Base vehicle for civilian agency deployments	GSA MAS application
NY State OGS	State-level AI governance	Centralized contract
State maritime / port authorities	Vessels — fleet visibility + sanctions screening	OGS / direct
State real-estate divisions	Terra — distressed-property and portfolio intel	OGS / direct

10. Verified Platform Metrics (re-verified 2026-05-05)

Metric	Value
Customer-facing product surfaces	8 + Alloy orchestration layer
Platform primitives	6
Database tables (provisioned)	848
API endpoint declarations	5,524
Industry verticals	7
Monorepo packages	126
Ouroboros packages	28
Original innovations (sovereign engine)	44
Ouroboros runtime test calls	133
Ouroboros guardrails tests	62
Codex v11 nodes / typed edges	76 / 95 across 11 domains
Security tests passing	126
CI workflows	23
Peer-style papers published	3 (v3, v9, v10)

11. What SZL Holdings Is Not Claiming

- Not currently performing on any federal contract
- No federal cloud authorization, no ATO, no DoD impact-level designation
- Not externally audited
- No signed customer contracts for the platform
- Single-founder operation
- Numbers above are internal platform metrics — not revenue, not users

The strength of the position is the public proof, not pretended traction.

12. 90-Day Roadmap

Timeline	Milestone
Days 1-14	SAM.gov UEI activated, primary NAICS confirmed
Days 15-30	CAGE issued, CMMC Level 1 self-assessment posted, first sources-sought scan pass
Days 31-60	First SBIR opportunity identified with APEX guidance, per-lane CPS payloads complete for Vessels + Terra
Days 61-90	First sources-sought response submitted, CPS rollout complete across remaining lanes , v11 thesis (planned) on CPS as a procurement protocol

13. Contact

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Attachments

- Operational Briefing v2 ([APEX_v2_Operational_Briefing.md](#))
- Investor Pitch Deck v2 ([APEX_v2_Investor_Pitch_Deck.md](#))
- Capability Statement v2 ([APEX_v2_Capability_Statement.md](#))
- Live Demo Guide v2 ([APEX_v2_Live_Demo_Guide.md](#))
- Meeting Script v2 ([APEX_v2_Meeting_Script.md](#))
- Audit Delta May 4 → May 5 ([APEX_v2_Audit_Delta_May5.md](#))
- Product screenshots
- Ouroboros Thesis v3, v9, v10 PDFs (Zenodo + GitHub Releases)